CO-Oxidation Proceeds via a Monodentate Carbonate-Facilitated Mechanism over Mixed $LaFe_{1-x}Co_xO_3$ Catalysts

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Introduction

CO oxidation is an apparently simple and extensively studied probe reaction over many catalysts and a Langmuir-Hinshelwood mechanism is well-established on noble metals.[1,2] Also transition metal oxides, such as $LaFeO_3$[3] or Co_3O_4[4] have been recently shown to be highly active, but their reaction mechanisms are less clear, in particular regarding a possible participation of surface carbonates. From a series of $LaFe_{1-x}Co_xO_3$ catalysts[3], we focus on two with x=0 and x=0.25 to unravel the roles of cobalt and of carbonates on CO oxidation by using a combination of *operando* NAP-XPS, DFT and DRIFTS-MES.

Experimental/methodology

Catalyst precursors were synthesized with the respective Fe:Co ratio via co-precipitation in an automated lab reactor (OptiMax 1001, Mettler Toledo, 10 °C, pH 9.5) and thermally decomposed to the perovskites in a muffle furnace at 600 °C in air. Steady State CO-oxidation experiments were carried out in a home-build setup (2% CO, 1% O_2 rest nitrogen, 50 mg catalyst, sieve fraction 250-355 µm). DRIFTS-MES was recorded on *a ThermoFisher Nicolet™ iS20* FTIR spectrometer with the *Praying Mantis™* accessory coupled with MS. For simulating the CO-Oxidation process, we adopted the density functional theory calculations with Habbard U term, implemented in CASTEP, the Cambridge Sequential Total Energy Package of Material Studio.[5,6] *Operando* NAP-XPS was recorded at the ISISS beamline at BESSY, Berlin at five different temperatures, each held for 2 hours.

Results and Discussion

The $LaFeO_3$ catalyst shows low activity despite a lower apparent activation energy (Fig. 1a). The reaction order in oxygen near zero suggests a Mars-van Krevelen type

mechanism (Fig. 1b). DFT calculations reveal that adsorbed CO binds to the oxide surface as a bidentate carbonate (step 3 in Fig. 1c), which is decomposes into CO_2 and an oxygen vacancy (step 4). This was experimentally verified by ME-DRIFTS and a phase delay analysis at higher demodulation indices (k >=3) clearly showed that the carbonates are formed first on the surface before CO_2 accumulation takes place in the gas-phase. The introduction of cobalt leads to a much higher rate and to higher reducibility of the surface. The increased reaction order in oxygen suggests increasing contribution of a Langmuir Hinshelwood type mechanism. This also changes the reactive carbonate to a monodentate species on $LaFe_{0.75}Co_{0.25}O_3$, which was observed via ME-DRIFTS and in calculations. This monodentate carbonate is converted faster according to our phase delay analysis. A carbonate-facilitated mechanism of CO_2 formation is therefore proposed for both catalysts, which is also in accordance with observed carbonaceous species during *operando* NAP-XPS. The presence of cobalt in $LaFe_{0.75}Co_{0.25}O_3$ leads to a change from bi- to more reactive monodentate surface carbonate as a precursor for CO_2.

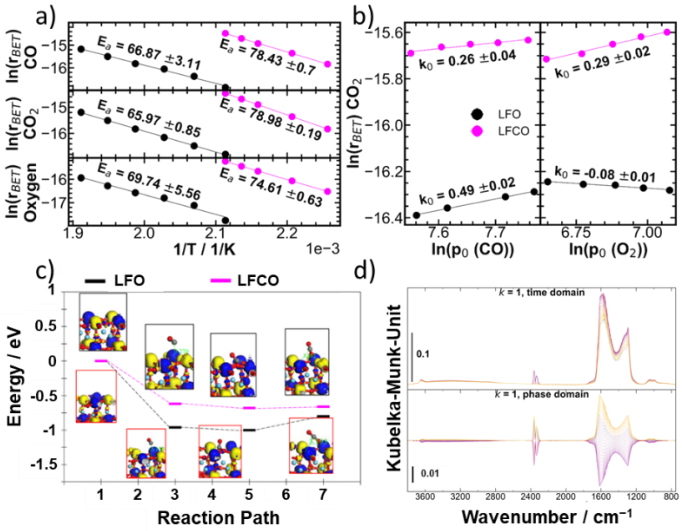

Figure 1: Steady State results showing Arrhenius (a) and power-rate law results (b); Calculated reaction pathways for both catalyst for a MvK mechanism with CO adsorbing on the surface(c); ME-DRIFTS spectra (d) of LFCO at 200 °C comparing the time and phase domain data.

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